STRUCTURED FORWARD–REVERSE FRAMEWORK FOR DYSARTHRIC SPEECH MODELING AND ENHANCEMENT

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ABSTRACT

Dysarthric speech poses significant challenges for automatic speech recognition (ASR) due to temporal elongation, articulatory imprecision, phonation instability, and irregular prosody. In this work, we propose a structured, text-free framework for modeling and reversing moderate dysarthria directly in self-supervised speech embedding space. Clean phoneme-level embeddings are first extracted using a pretrained encoder and then corrupted through controlled modules that simulate clinically observed distortions, including probabilistic time-warping, articulation smoothing, phonation perturbation, and pause–rhythm variability, while preserving monotonic alignment and linguistic structure. To reconstruct clean representations, we design a factorized reverse architecture with dedicated correction heads and an adaptive gating mechanism to fuse distortion-specific corrections. The reconstructed embeddings are converted to waveform using a neural vocoder, and performance is evaluated through embedding similarity and downstream ASR improvement. This framework provides a modular and interpretable approach to dysarthria modeling and intelligibility restoration without reliance on text-aligned supervision.

1 INTRODUCTION

Dysarthria is a motor speech disorder resulting from neurological impairments that affect the strength, coordination, or timing of the speech musculature. These deficits manifest acoustically as temporal elongation, articulatory imprecision, unstable phonation, irregular pauses, and prosodic abnormalities Duffy (2013); Yorkston et al. (1999). As a consequence, automatic speech recognition (ASR) systems trained on typical speech exhibit severe performance degradation when applied to dysarthric speech Rudzicz (2013); Christensen et al. (2012). This degradation limits the usability of modern voice-driven technologies for individuals with motor speech disorders, raising both accessibility and equity concerns.

Existing approaches for improving dysarthric speech recognition broadly fall into two categories: (i) adaptation-based ASR techniques, which fine-tune acoustic models using dysarthric data Christensen et al. (2010), and (ii) speech enhancement or conversion frameworks that attempt to transform dysarthric speech into a more intelligible form prior to recognition Toda et al. (2007); Qian et al. (2014). While effective in controlled settings, many of these methods rely on parallel clean–dysarthric recordings or explicit text supervision. Such requirements are difficult to satisfy in practice, particularly for low-resource languages and clinical populations where labeled data is scarce. Furthermore, paired data collection is expensive and often infeasible for speakers with severe motor impairments.

This work addresses the following research question: Can moderate dysarthric speech be modeled and reconstructed in a structured, text-free manner directly in self-supervised speech

embedding space, while preserving linguistic content and improving downstream ASR performance?

Rather than operating at the waveform level, we propose to model dysarthric characteristics in phoneme-level representations extracted using self-supervised learning (SSL) encoders such as wav2vec 2.0 Baevski et al. (2020b). SSL models learn robust linguistic abstractions from large-scale unlabeled speech and have demonstrated strong transferability across tasks and domains Baevski et al. (2020b); Schneider et al. (2019). By working in embedding space, we can introduce controlled distortions that mimic clinically observed dysarthric features while maintaining monotonic alignment and semantic structure.

Specifically, we construct a structured forward corruption process that simulates moderate dysarthria through: (i) stochastic temporal elongation using probabilistic time-warping, (ii) articulation smoothing to emulate reduced articulatory precision, (iii) phonation instability to model breathiness and voicing variability, and (iv) pause–rhythm perturbations to capture irregular timing patterns. These modules are designed to reflect acoustic characteristics documented in clinical and phonetic literature [1], [2], while remaining statistically measurable and controllable.

To reverse these distortions, we introduce a factorized reconstruction architecture consisting of a shared backbone with specialized correction heads corresponding to each corruption type. An adaptive gating mechanism learns to fuse distortion-specific corrections, yielding reconstructed clean embeddings that are subsequently converted to waveform using a neural vocoder Kong et al. (2020b). Unlike traditional enhancement approaches, this framework does not require parallel clean supervision at the waveform or transcript level, enabling scalable modeling in low-resource settings.

By explicitly decomposing dysarthric variability into interpretable components and modeling them in embedding space, this work contributes a modular and structured framework for intelligibility restoration. Beyond ASR improvement, such modeling has implications for assistive communication technologies and neuro-speech interfaces, where text-independent reconstruction may be critical.

2 DATASET USED

This study also employs the TORGO database Rudzicz et al. (2012), which contains speech recordings from 8 speakers with dysarthria (3 females, 5 males) and 7 control speakers (3 females, 4 males). The dysarthric speakers exhibit symptoms of either cerebral palsy or amyotrophic lateral sclerosis (ALS), with varying degrees of speech impairment, providing a diverse sample of articulatory deviations. The TORGO corpus includes both isolated words and read sentences, recorded in a controlled acoustic environment using a head-mounted microphone and an electromagnetic articulograph (EMA) setup, with audio sampled at 16 kHz.

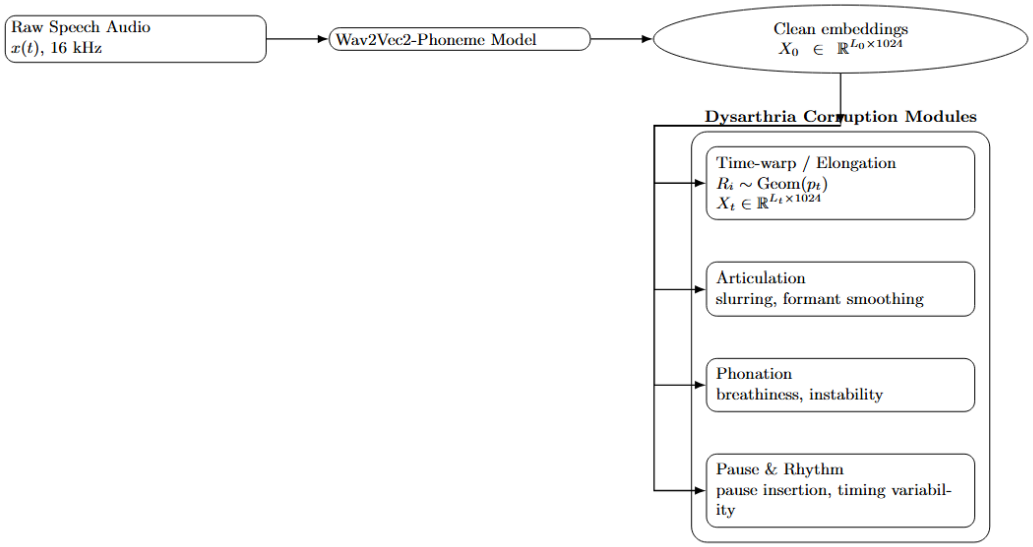
In this study, only the audio recordings captured using the head-mounted microphone are utilized to ensure consistency with the other datasets and compatibility with pretrained self-supervised models. All available speech samples from both dysarthric and control speakers are included to maximize data availability, speaker diversity, and phonetic coverage. However, the primary modeling focus is placed on speakers exhibiting moderate dysarthria as shown in Table 1. Moderate severity provides an appropriate balance between distortion and recoverability: while the speech contains clear and measurable temporal elongation, articulatory imprecision, and prosodic instability, the underlying linguistic structure remains largely intact. In contrast, mild dysarthria often exhibits subtle deviations that are difficult to model explicitly, whereas severe dysarthria may introduce excessive degradation that compromises stable alignment and reconstruction.

Severity	Speaker	Final utterances
Mild	F04	100
	M03	95
Moderate	F03	138
	M05	124
Severe	F01	20
	M01	84
	M02	91
	M04	84
Total		**736**

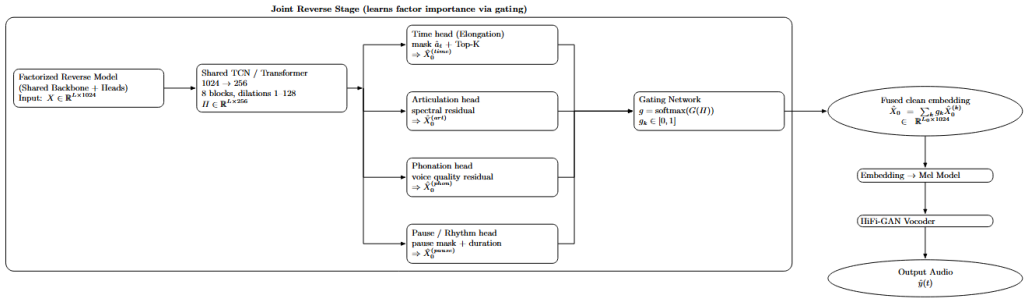
Table 1: Final utterance count per speaker and severity level

3 METHODOLOGY

3.1 ARCHITECTURE OVERVIEW



(a) Forward stage: embedding extraction and structured dysarthria corruption modules.



(b) Reverse stage: factorized reconstruction heads with gating and waveform synthesis.

Figure 1: Proposed embedding-space dysarthria modeling framework: forward corruption and factorized reverse reconstruction.

Our framework models dysarthria as a structured corruption process in self-supervised phoneme-level embedding space, followed by a factorized reverse model that reconstructs clean embeddings. Given raw speech audio $x(t)$ sampled at 16 kHz, we first extract clean embeddings using a pretrained Wav2vec2 phoneme-level encoder Baevski et al. (2021):

$$X_0 = f_{\text{enc}}(x(t)), \qquad X_0 \in \mathbb{R}^{L_0 \times 1024}, \tag{1}$$

where L_0 denotes the original sequence length and 1024 is the embedding dimension.

3.1.1 Forward Dysarthria Corruption (Structured Modeling)

We simulate moderate dysarthric characteristics using a set of modular corruption operators applied in embedding space. Starting from X_0, we obtain a corrupted sequence X_t:

$$X_t = \mathcal{C}(X_0) = \mathcal{C}_{\text{rhythm}} \circ \mathcal{C}_{\text{phon}} \circ \mathcal{C}_{\text{art}} \circ \mathcal{C}_{\text{time}}(X_0), \tag{2}$$

where each component targets a clinically motivated distortion: (i) time-warp/elongation, (ii) articulation (e.g., slurring / formant smoothing), (iii) phonation (e.g., breathiness / instability), and (iv) pause–rhythm variability.

For temporal elongation, we apply probabilistic time-warping by repeating frames according to a geometric run-length model. Let $R_i \sim \text{Geom}(p_t)$ be the number of repeats for frame i, bounded by a maximum run length $L_{\max}$ for stability. This produces a monotonic expansion of the sequence length from L_0 to L_t while preserving frame order:

$$X_t^{(\text{time})} \in \mathbb{R}^{L_t \times 1024}, \qquad L_t \geq L_0. \tag{3}$$

3.1.2 Reverse Modeling (Factorized Reconstruction + Gating)

To reconstruct clean embeddings from the corrupted representation X_t, we employ a factorized reverse model with a shared backbone and distortion-specific heads. A shared temporal backbone (TCN/Transformer) first maps the input embeddings into a hidden representation:

$$H = f_{\text{backbone}}(X_t), \qquad H \in \mathbb{R}^{L \times 256}. \tag{4}$$

We then predict multiple candidate reconstructions using specialized heads, each focusing on undoing one distortion factor:

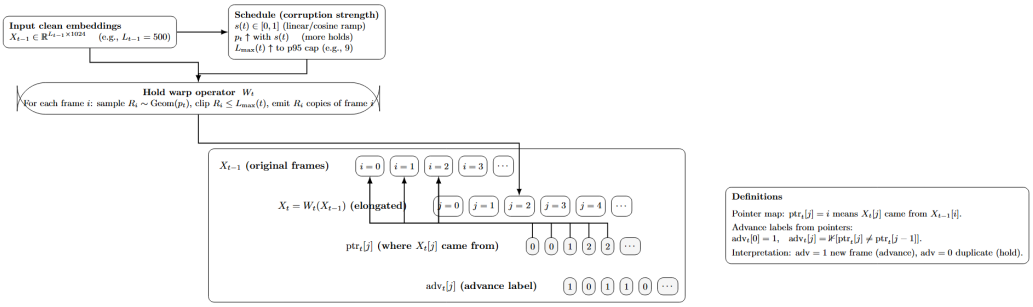
$$\hat{X}_0^{(k)} = h_k(H), \qquad k \in \{\text{time, art, phon, rhythm}\}. \tag{5}$$

A gating network computes importance weights for each factor and fuses the candidate reconstructions:

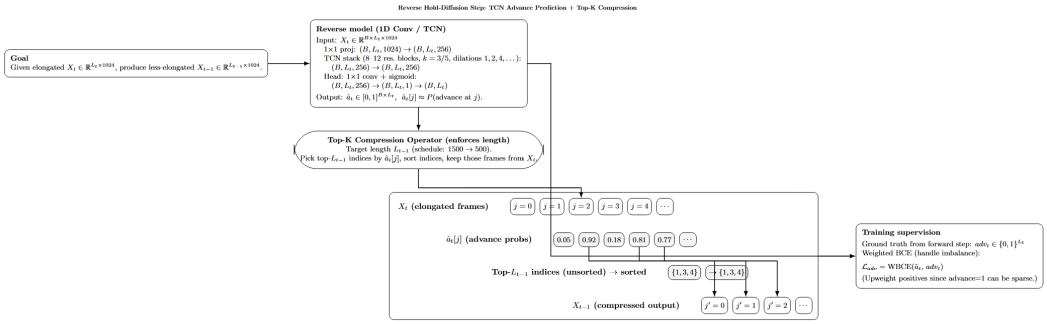
$$g = \text{softmax}(G(H)), \qquad \hat{X}_0 = \sum_k g_k \, \hat{X}_0^{(k)}, \qquad \hat{X}_0 \in \mathbb{R}^{L_0 \times 1024}. \tag{6}$$

Finally, the reconstructed embeddings $\hat{X}_0$ are converted to waveform using an embedding-to-mel model followed by a HiFi-GAN vocoder to obtain enhanced speech $\hat{y}(t)$.

3.2 TEMPORAL ELONGATION VIA HOLD-WARP (PTR)



(a) Forward hold-warp operator W_t: geometric frame holding with pointer map ptr_t and advance labels adv_t.



(b) Reverse hold-diffusion step: TCN-based advance prediction $\hat{a}_t$ followed by Top-K monotonic compression to produce X_{t-1}.

Figure 2: Temporal elongation modeling and reversal in embedding space.

A prominent characteristic of moderate dysarthria is temporal elongation, where speakers hold phones longer and exhibit slowed transitions. We model this effect directly in embedding space using a stochastic *hold-warp* operator that repeats (holds) frames while preserving monotonic order.

Let $X_{t-1} \in \mathbb{R}^{L_{t-1} \times 1024}$ denote the input embedding sequence at step $t-1$. The forward elongation operator W_t produces an elongated sequence $X_t \in \mathbb{R}^{L_t \times 1024}$ by sampling a run length for each frame:

$$R_i \sim \text{Geom}(p_t), \qquad R_i \leftarrow \min(R_i, L_{\max}(t)), \tag{7}$$

and emitting R_i copies of frame $X_{t-1}[i]$. This yields:

$$X_t = W_t(X_{t-1}), \qquad L_t = \sum_{i=0}^{L_{t-1}-1} R_i. \tag{8}$$

A schedule $s(t) \in [0, 1]$ controls corruption strength across steps by increasing the expected holds (via p_t) and optionally adjusting the cap $L_{\max}(t)$.

Pointer map and advance labels. During forward warping we also record a pointer map ptr_t where $\text{ptr}_t[j] = i$ indicates that $X_t[j]$ originated from $X_{t-1}[i]$. From pointers, we derive binary advance labels $\text{adv}_t \in \{0, 1\}^{L_t}$:

$$\text{adv}_t[0] = 1, \qquad \text{adv}_t[j] = \mathbb{I}\left[\text{ptr}_t[j] \neq \text{ptr}_t[j-1]\right]. \tag{9}$$

Intuitively, $\text{adv}_t[j] = 1$ marks a transition to a new original frame (advance), while $\text{adv}_t[j] = 0$ corresponds to a duplicated (held) frame.

3.3 Reverse Hold-Diffusion Step (Advance Prediction + Compression)

To reverse elongation, we train a 1D temporal model (TCN / 1D Conv) that predicts an *advance probability* for each elongated position:

$$\hat{a}_t[j] \approx \mathbb{P}(\mathrm{adv}_t[j] = 1 \mid X_t), \qquad \hat{a}_t \in [0,1]^{L_t}. \tag{10}$$

Given a target length L_{t-1} (scheduled from longer to shorter sequences), we apply a Top-K compression operator that selects the $K = L_{t-1}$ indices with the highest $\hat{a}_t[j]$, sorts them, and keeps the corresponding frames from X_t to form the compressed output X_{t-1}. The model is trained using weighted binary cross-entropy to handle class imbalance, using the forward-derived labels adv_t as supervision:

$$\mathcal{L}_{\mathrm{adv}} = \mathrm{WBCE}(\hat{a}_t, \mathrm{adv}_t). \tag{11}$$

This formulation enforces monotonic compression while learning to identify true frame transitions, thereby undoing hold-induced elongation in a controlled and interpretable manner.

3.4 Embedding to Mel Spectrogram to Waveform Conversion

To convert wav2vec 2.0 Baevski et al. (2020b) representations to mel spectrograms, we train a 1D Conv model using embeddings as inputs and true mel spectrograms as outputs, as shown in Figure 4. As consecutive time frames in the embedding space are spaced by 20 ms, we maintain this layout for mel spectrograms. Upon model training, mel spectrograms were converted to 16000 Hz audio waveforms using the hifi-gan model by Kong et al. (2020a) for human interpretation.

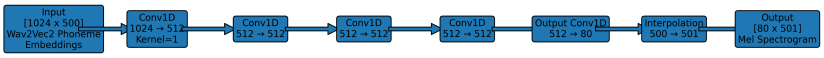


Figure 3: CNN model architecture for conversion of wav2vec2phoneme embeddings to Mel spectrogram.

3.5 Combined System Architecture

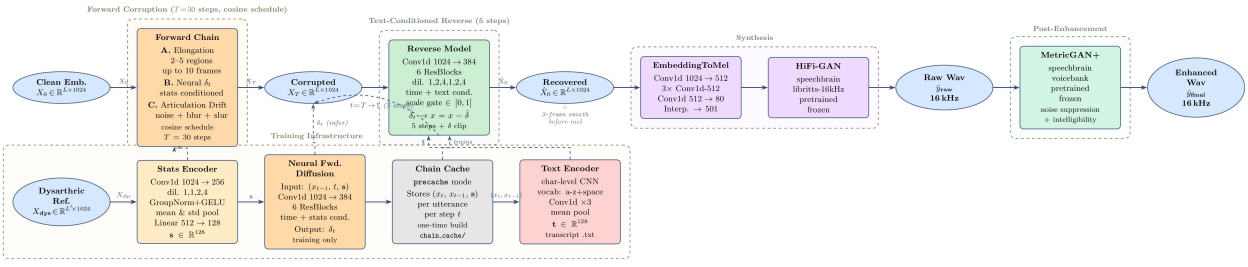


Figure 4: Overall Architecture Diagram of the Reconstruction Pipeline

The proposed system as shown in Fig 4 consists of two coupled pipelines: a forward simulation pipeline that corrupts clean speech embeddings to resemble dysarthric speech, and a reverse remediation pipeline that attempts to recover intelligibility from dysarthric embeddings. Both pipelines operate entirely in the HuBERT/wav2vec2 embedding space (dimensionality 1024) rather than on raw waveforms or mel spectrograms, which provides a compact, phonetically rich representation that abstracts away speaker-level acoustic variation.

3.5.1 STATS ENCODER

The StatsEncoder is a 1D convolutional network that takes a dysarthric speech embedding sequence $X_{dys} \in \mathbb{R}^{L\times 1024}$ and produces a compact 128-dimensional conditioning vector $\mathbf{s}$ that captures the statistical characteristics of the dysarthric speaker's embedding distribution.

The architecture consists of four dilated convolutional layers with kernel sizes of 7, 5, 5, and 5 and dilation rates of 1, 1, 2, and 4 respectively, each followed by GroupNorm and GELU activation. After the convolutional stack, global mean and standard deviation pooling are applied across the time dimension, producing two 256-dimensional vectors that are concatenated and projected to 128 dimensions via a linear layer. The output $\mathbf{s} \in \mathbb{R}^{128}$ encodes both the mean activation pattern and the variance of the dysarthric embedding — effectively capturing how far the dysarthric speaker's phoneme realisations deviate from typical clean speech in embedding space.

This vector serves as the conditioning signal for the NeuralForwardDiffusion model. Crucially, since the TORGO dataset has no paired data, the encoder is trained using a proxy objective: given a dysarthric embedding, predict the mean and standard deviation statistics of a randomly associated clean embedding. This forces the encoder to learn which directions in embedding space correspond to dysarthric degradation, without requiring utterance-level pairing.

3.5.2 NEURAL FORWARD DIFFUSION

The NeuralForwardDiffusion model learns to predict the corruption delta δ_t at each diffusion step t, conditioned on the current embedding state $\mathbf{x}_{t-1}$, the step index t, and the dysarthric style vector $\mathbf{s}$.

The architecture follows a 1D temporal convolutional network (TCN) design. The input embedding is projected from 1024 to 384 channels via a pointwise convolution. Six residual blocks with kernel size 5 and dilation rates 1, 2, 4, 1, 2, 4 then process the projected sequence. Each residual block applies two dilated convolutions separated by GroupNorm and GELU activations, with a skip connection bypassing the block. The receptive field of this stack covers approximately 60 frames, sufficient to capture phoneme-level temporal context.

The step index t is encoded via a sinusoidal time embedding of dimension 128, which is then projected to 384 dimensions via a two-layer MLP. Similarly, the style vector $\mathbf{s}$ is projected to 384 dimensions via a two-layer MLP. These two conditioning signals are summed and broadcast as a constant bias across all time positions before the first residual block. This additive conditioning mechanism allows the model to globally shift the feature representation based on both how far along the diffusion chain we are and what kind of dysarthric degradation is being simulated.

The output projection maps the 384-dimensional hidden state back to 1024 dimensions, producing the predicted delta δ_t. The corrupted state is then updated as $\mathbf{x}_t = \mathbf{x}_{t-1} + \alpha_t \cdot \delta_t$, where $\alpha_t = 0.5 \cdot sw_t$ and sw_t is the cosine schedule weight at step t.

The model is trained using an L1 + L2 combined loss comparing the predicted delta to the ground-truth delta derived from the pre-cached diffusion chain. The forward chain cache stores all 30 intermediate states for each utterance, so training reads directly from disk rather than recomputing chains on the fly.

3.5.3 ARTICULATION DRIFT (WITHIN THE FORWARD CHAIN)

At each forward step, in addition to the neural delta from Stage 2, an articulation drift perturbation is applied to $\mathbf{x}_t$. This consists of three sub-effects applied sequentially.

First, a smooth directional noise component is added. Unit-norm noise vectors are generated per frame and smoothed with an 11-frame moving average kernel to produce low-frequency directional drift. The magnitude of this drift is sampled from a truncated normal distribution scaled by $0.06 \cdot sw_t$, ensuring the effect is mild at early steps and grows gradually. The drift is renormalised so its per-frame magnitude stays bounded.

Second, a boundary blur is applied by blending the current embedding with a 5-frame moving average of itself, with a blend weight of $0.05 \cdot sw_t$. This simulates the smoothing of phoneme boundaries that occurs in dysarthric speech where transitions between phonemes are imprecise.

Third, a neighbour frame mixing step randomly selects a small fraction of frames (probability $0.04 \cdot sw_t$) and replaces them with a weighted average of the frame and its immediate neighbour, with mixing weight $0.05 \cdot sw_t$. This simulates the temporal slurring where adjacent phonemes blend into each other.

The three effects are intentionally mild — total scale 0.06 — so that elongation (Stage 2 of the forward chain) dominates the perceptual character of the simulated dysarthria, and articulation drift contributes a secondary spectral degradation on top of the temporal effects.

3.5.4 TEXT ENCODER

The TextEncoder is a character-level convolutional network that maps a transcript string to a 128-dimensional conditioning vector $\mathbf{t}$ for use in the reverse model.

Input characters are lowercased and mapped to integer indices over a vocabulary of 28 symbols (a–z, space, padding). Each character is embedded into a 64-dimensional vector via a learned embedding table. Three 1D convolutional layers with hidden dimension 256 and kernel size 3 process the embedded sequence, each followed by GELU activation. Global mean pooling across the character sequence produces a single 256-dimensional vector, which is projected to 128 dimensions. The encoder is trained jointly with the reverse model rather than separately.

The motivation for including text conditioning is fundamental to the remediation task. Without knowing what words were spoken, the reverse model has no phonetic anchor — it can only push the corrupted embedding toward the average clean speech distribution, which produces output that sounds like clean but generic speech rather than the specific utterance. With the transcript, the model can associate each region of the embedding with specific phonemes and guide the denoising toward the correct target.

3.5.5 REVERSEARTICULATIONUNETLITE

The reverse model takes a corrupted embedding $\mathbf{x}_t$, step index t, total steps T, and text conditioning vector $\mathbf{t}$, and predicts a gated correction delta $\hat{\delta}_t$ to subtract from $\mathbf{x}_t$.

The architecture mirrors the forward model in its TCN backbone: a pointwise projection from 1024 to 384, followed by six residual blocks with dilations 1, 2, 4, 1, 2, 4. The time embedding (sinusoidal, 128-dim, projected to 384) and text conditioning vector (projected from 128 to 384 via a two-layer MLP) are summed and added as a broadcast bias, conditioning all frames simultaneously on both the diffusion step and the phonetic content of the utterance.

The key architectural addition over the forward model is the scale gate, a learned per-sample scalar in $[0, 1]$ produced by a small MLP applied to the global mean of the hidden state. This gate multiplies the raw predicted delta before output: $\hat{\delta}_t = \text{gate} \cdot \delta_{\text{raw}}$. The gate acts as a learned confidence measure — when the model is uncertain about how much correction to apply (e.g., for very corrupted inputs where the embedding has drifted far from anything in training), the gate automatically reduces the magnitude of the correction. Without this mechanism, the raw delta can be large and lead to noise explosion across the 5 reverse steps.

At inference, a hard delta clip is additionally applied: $\hat{\delta}_t \leftarrow \text{clip}(\hat{\delta}_t,\ -0.20\|\mathbf{x}_t\|,\ +0.20\|\mathbf{x}_t\|)$, ensuring no single reverse step moves the embedding by more than 20% of its current norm. This provides a floor-level guarantee against runaway corrections that the soft scale gate alone may not prevent.

The training objective combines three losses. The primary loss regresses the predicted delta against the ground-truth delta $(x_t - x_{t-1})$ from the cached chain. A secondary loss compares the predicted previous state $\hat{x}_{t-1} = x_t - \hat{\delta}_t$ against the ground-truth x_{t-1}, keeping the full embedding trajectory grounded. A smoothness loss penalises frame-to-frame variation in $\hat{x}_{t-1}$, directly discouraging the jitter that manifests as high-frequency noise after vocoding. The three terms are weighted 1.0, 0.5, and 0.3 respectively. A cosine annealing learning rate schedule is used across epochs.

Inference uses 5 evenly-spaced reverse steps (e.g., $t = 30, 24, 18, 12, 6$) rather than all 30. Using all 30 steps compounds the per-step prediction error — each step introduces a small approximation

error that accumulates multiplicatively. With 5 steps the total accumulated error is bounded and the recovered embedding remains close to the clean manifold.

3.5.6 POST-PROCESSING

After the reverse model outputs $\hat{x}_0$, two post-processing steps are applied before synthesis.

First, a 3-frame temporal smoothing is applied to $\hat{x}_0$ using average pooling with a kernel size of 3 and padding of 1. This removes residual frame-to-frame jitter in the recovered embedding, small per-frame oscillations that are invisible in the embedding space but cause HiFi-GAN to produce audible high-frequency artefacts.

Second, the smoothed embedding is passed through EmbeddingToMel, a 1D convolutional decoder that maps the 1024-dimensional embedding to an 80-bin mel spectrogram of 501 frames. The architecture consists of a pointwise projection from 1024 to 512, three convolutional layers with kernel size 5 and 512 channels each (with BatchNorm and ReLU), a final projection to 80 channels, and linear interpolation to the target length. HiFi-GAN (SpeechBrain's libritts-16kHz pretrained model) then converts the mel spectrogram to a 16 kHz waveform.

Finally, MetricGAN+ is applied as a standalone post-enhancement stage. MetricGAN+ is a pretrained GAN trained on VoiceBank-DEMAND using a discriminator that directly targets PESQ improvement. It requires no fine-tuning and processes the recovered waveform to suppress the harmonic artefacts introduced by HiFi-GAN when given an embedding that deviates from its clean-speech training distribution. In our evaluation, MetricGAN+ improved STOI by approximately 0.08 on average and subjectively removed the high-frequency noise that was the primary complaint about intelligibility.

4 RESULTS

4.1 INTELLIGIBILITY AND PERCEPTION BASED SCORES

The STOI and PESQ scores were calculated for reconstructed speech with respect to the input dysarthric speech, as shown in Table 2. The STOI score of 0.864 indicates that the pipeline successfully enhances dysarthric speech. On the other hand, the PESQ score of 2.829 indicates that some aspects of speech quality may be affected by distortions resulting from conversion of embeddings to mel spectrograms to waveforms.

STOI	**PESQ**
0.864	2.829

Table 2: STOI and PESQ scores for reconstructed speech with respect to input dysarthric speech

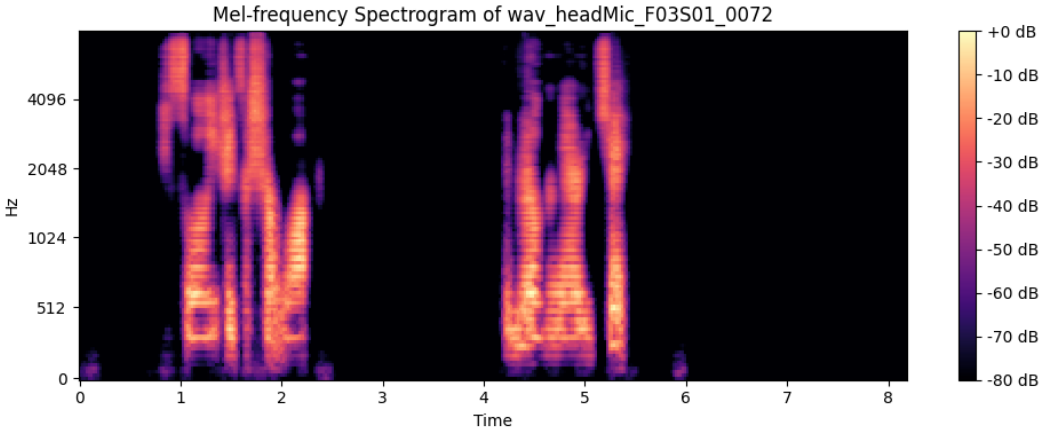
4.2 ASR BASED METRICS

The WER and CER scores were calculated for both dysarthric and reconstructed speech using the wav2vec 2.0 model by Baevski et al. (2020a), as shown in Table 3. While intelligibility and perception based scores were promising, ASR based metrics suffer primarily due to the presence of distortion artifacts in the audio. Furthermore, we hypothesize that noise from both forward and reverse diffusion may contribute to poorer ASR performance in reconstructed speech.

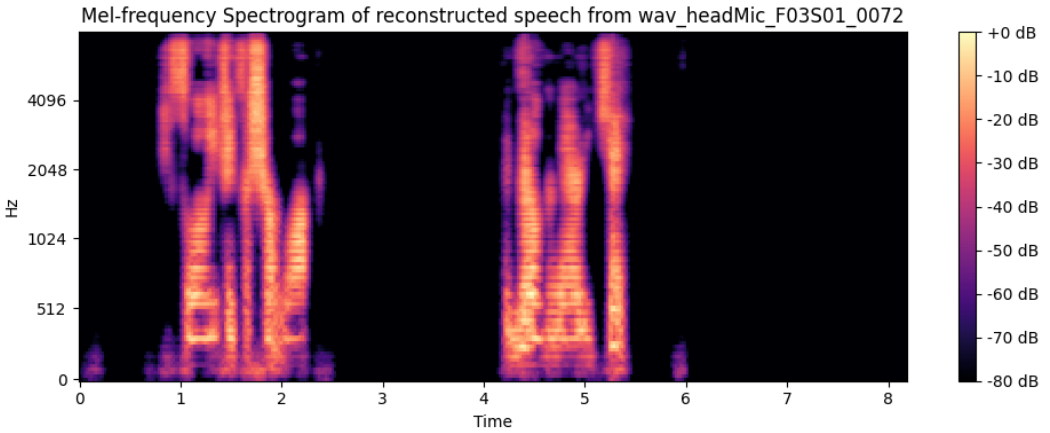
Metric	**Dysarthric**	**Reconstructed**
WER	0.928	0.943
CER	0.615	0.639

Table 3: WER and CER metrics for dysarthric input speech and reconstructed speech

4.3 MEL SPECTROGRAMS OF INPUT VS RECONSTRUCTED SPEECH



(a) Mel Spectrogram of a dysarthric speech input example



(b) Mel Spectrogram of a reconstructed speech example

Figure 5: Comparison of Mel Spectrograms of input (top) and reconstructed audio (bottom)

Mel Spectrograms of an input recording of moderate dysarthria and of the reconstruction are shown in Figure 5. The plots demonstrate that the energy distribution is consistent between the two.

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